

Problem 1

Given in Professor C's lecture notes we can now start at equation one:

$$1.) dH = C_p dT + (1 - \beta T) V dP$$

Since we are dealing with an adiabatic expansion there is no change in enthalpy therefore $dH = 0$

$$2.) 0 = C_p dT + (1 - \beta T) V dP$$

$$3.) -(1 - \beta T) = C_p \left(\frac{\partial T}{\partial P}\right)_H \left(\frac{1}{V}\right)$$

Given in lecture we know the equation for density and the expansivity constant.

$$4.) \frac{1}{V} = \rho$$

$$5.) \beta = \left(\frac{1}{V}\right) \left(\frac{\partial V}{\partial T}\right)_P$$

We can now plug in the equation of the expansivity constant and can plug it into equation 6

$$6.) -1 + \left(\frac{T}{V}\right) \left(\frac{\partial V}{\partial T}\right)_P = C_P \left(\frac{\partial T}{\partial P}\right)_H \rho$$

We can now separate the $\left(\frac{T}{V}\right) \left(\frac{\partial V}{\partial T}\right)_P$

$$7.) -1 + \left(\frac{T}{\partial T}\right)_P * \left(\frac{\partial V}{V}\right)_P = C_P \left(\frac{\partial T}{\partial P}\right)_H \rho$$

We must now integrate equation (7) where the integral of $\left(\frac{\partial V}{V}\right)_P = \ln(V)$ and the integral of $\left(\frac{T}{\partial T}\right)_P = 1/\ln(T)$

$$8.) -1 + \int \left(\frac{T}{\partial T}\right)_P * \int \left(\frac{\partial V}{V}\right)_P = C_P \left(\frac{\partial T}{\partial P}\right)_H \rho$$

$$9.) -1 + \left(\int \left(\frac{T}{\partial T}\right)_P\right)' * \left(\int \left(\frac{\partial V}{V}\right)_P\right)' = C_P \left(\frac{\partial T}{\partial P}\right)_H \rho$$

Taking the derivative of the integral in part (8) we get $\partial \ln(V)$ and $\frac{1}{\partial \ln(T)}$

$$10) -1 + \left(\frac{\partial \ln(V)}{\partial \ln(T)} \right)_P = C_P \left(\frac{\partial T}{\partial P} \right)_H \rho$$

To calculate the temperature change of an ideal gas that is undergoing an adiabatic expansion through a throttling Valve

$$11) Q = 0, W = 0, H_1 = H_2, \Delta H = 0$$

$$12) dH = C_p dT + (1 - \beta T) V dP$$

$$13) dH = C_p dT$$

We rearrange the formula to solve for the specific heat

$$14) C_p = \frac{\partial H}{\partial T}$$

$$15) \frac{\partial H}{\partial T} * \rho * \left(\frac{dT}{dP} \right)_H = \frac{\partial \ln(V)}{\partial \ln(T)} - 1$$

$$16) \frac{\partial \ln(V)}{\partial \ln(T)} = 1$$

WE must now integrate in order to separate the temperature values

$$17) \int_{V_1}^{V_2} d \ln V = \int_{T_1}^{T_2} d \ln T$$

$$18) \ln(V) \big|_{V_1}^{V_2} = \ln(T) \big|_{T_1}^{T_2}$$

$$19) \ln(V_2) - \ln(V_1) = \ln(T_2) - \ln(T_1)$$

Once we have integrated we must get rid of the natural log by raising the expression by e

$$20) \ln \left(\frac{V_2}{V_1} \right) = \ln \left(\frac{T_2}{T_1} \right)$$

$$21) e^{\ln \left(\frac{V_2}{V_1} \right)} = e^{\ln \left(\frac{T_2}{T_1} \right)}$$

$$22) \left(\frac{V_2}{V_1} \right) = \left(\frac{T_2}{T_1} \right)$$

We can now separate the expression to solve for one T value

$$23) \left(\frac{V_2 * T_1}{V_1} \right) = T_2$$

We can subtract T_1 from both sides now in order to acquire the change in temperature.

$$24) \left(\frac{V_2 * T_1}{V_1} \right) - T_1 = T_2 - T_1$$

$$25) \Delta T = \left(\frac{V_2 * T_1}{V_1} \right) - T_1$$

Problem 2

Given in Professor C's lecture notes we can now start at equation one:

$$1) dH = C_p dT + (1 - \beta t) V dP$$

$$2) dU = C_v dT + \left(\frac{\partial U}{\partial V} \right)_T dV$$

with constant pressure we get

$$3) \text{ constant } P: dH = C_p dT$$

With constant volume we get

$$4) \text{ constant } V: dU = C_v dT$$

$$5) C_v \left(\frac{\partial P}{\partial \rho} \right)_S = C_p \left(\frac{\partial P}{\partial \rho} \right)_T$$

$$6) C_v \left(\frac{\partial P}{\partial V} \right)_S = C_p \left(\frac{\partial P}{\partial V} \right)_T$$

$$7) d\rho = -\frac{1}{V^2}$$

We must prove :

$$8) \left(\frac{\partial P}{\partial V} \right)_S = -\left(\frac{\partial H}{\partial V} \right)_P / \left(\frac{\partial U}{\partial P} \right)_V$$

$$9) \left(\frac{\partial H}{\partial V} \right)_P = C_p \left(\frac{\partial T}{\partial V} \right)_P$$

$$10) \left(\frac{\partial U}{\partial P} \right)_V = C_v \left(\frac{\partial T}{\partial P} \right)_V$$

WE can take the total differential of temperature with respect to pressure and volume

$$11) T = T(P, V) = dT = \left(\frac{\partial T}{\partial P} \right)_V dP + \left(\frac{\partial T}{\partial V} \right)_P dV$$

$$12) \left(\frac{\partial T}{\partial V} \right) = \left(\frac{\partial T}{\partial p} \right)_V \left(\frac{\partial P}{\partial V} \right) + \left(\frac{\partial T}{\partial v} \right)_P \left(\frac{\partial V}{\partial V} \right)$$

$$13) \left(\frac{\partial P}{\partial V}\right)_T = -\left(\frac{\partial T}{\partial V}\right)_P / \left(\frac{\partial T}{\partial P}\right)_V$$

$$14) \left(\frac{\partial P}{\partial V}\right)_S = -\left(\frac{\partial H}{\partial V}\right)_P / \left(\frac{\partial U}{\partial P}\right)_V$$

$$15) dU = TdS - PdV$$

$$16) U = U(P, V)$$

We can relate Both internal energy functions

$$17) dU = \left(\frac{\partial U}{\partial P}\right)_V dP + \left(\frac{\partial U}{\partial V}\right)_P dV = TdS - PdV$$

$$18) T \left(\frac{\partial S}{\partial V}\right) = \left(\frac{\partial U}{\partial P}\right)_V \left(\frac{\partial P}{\partial V}\right) + \left[\left(\frac{\partial U}{\partial V}\right)_P + P \left(\frac{\partial V}{\partial V}\right)\right]$$

$$19) \left(\frac{\partial U}{\partial P}\right)_V * \left(\frac{\partial P}{\partial V}\right)_S = -[P + \left(\frac{\partial U}{\partial V}\right)_P]$$

$$20) H = U + PV$$

We must now take the total differential of enthalpy

$$21) dH = dU + PdV + VdP$$

$$22) \left(\frac{\partial H}{\partial V}\right)_P = \left(\frac{\partial U}{\partial V}\right)_P + P$$

$$23) \left(\frac{\partial U}{\partial P}\right)_V \left(\frac{\partial P}{\partial V}\right)_S = [-P + \left(\frac{\partial U}{\partial V}\right)_P]$$

$$24) \left(\frac{\partial P}{\partial V}\right)_S = [P + \left(\frac{\partial U}{\partial V}\right)_P] / \left(\frac{\partial U}{\partial P}\right)_V$$

$$25) [P + \left(\frac{\partial U}{\partial V}\right)_P] / \left(\frac{\partial U}{\partial P}\right)_V = -\left(\frac{\partial H}{\partial V}\right)_P / \left(\frac{\partial U}{\partial P}\right)_V$$

$$26) -[\left(\frac{\partial U}{\partial V}\right)_P + P] / \left(\frac{\partial U}{\partial P}\right)_V = \frac{\partial P}{\partial V}_S$$

Calculate the adiabatic speed of sound:

$$1) C_V \left(\frac{\partial P}{\partial \rho}\right)_S = C_P \left(\frac{\partial P}{\partial \rho}\right)_T$$

$$2) V^2 = \left(\frac{\partial P}{\partial \rho}\right)_S$$

$$3)V^2 = C_p \left(\frac{\partial P}{\partial \rho} \right)_T$$

The following specific heat values were obtained from the engineering toolbox at 25°C

$$4)C_p = 1.006 \left(\frac{kJ}{kg K} \right)$$

$$5)C_v = .7175 \left(\frac{kJ}{kg K} \right)$$

$$6)T = 298.15 K, R = 287 \left(\frac{J}{kg K} \right)$$

We must now rearrange the ideal gas equation

$$7)PV = RT$$

$$8) \left(\frac{P}{\rho} \right) = RT$$

$$9)P = \rho RT$$

$$10) \left(\frac{\partial P}{\partial \rho} \right)_T = \left(\frac{dPRT}{d\rho} \right)_T$$

$$11) \left(\frac{\partial P}{\partial \rho} \right)_T = RT$$

We must now insert equation (11) into equation (3)

$$12)V^2 = C_p/C_v(RT)$$

The values I acquired from online(engineering toolbox) was $C_p = 1.006 \left(\frac{kJ}{kmol} \right)$ and $C_v = .7175 \left(\frac{kJ}{kmol} \right)$ at 25°C and the value of R is $287 \left(\frac{J}{kgK} \right)$

$$13) V^2 = \left(\frac{1.006 \left(\frac{kJ}{kgK} \right)}{.7175 \left(\frac{kJ}{kgK} \right)} \right) * 287 \left(\frac{J}{kgK} \right) * (298.15K)$$

$$14) V^2 = 119975.56 \left(\frac{J}{kg} \right) * \left(\frac{kg * m^2}{J * s^2} \right)$$

$$15) V^2 = 119975.56 \left(\frac{m^2}{s^2} \right)$$

$$16) V = \sqrt{(119975.56) \left(\frac{m^2}{s^2} \right)}$$

$$17) v = 346 \text{ m/s}$$

Problem 3

#3 Part(1)

In order to calculate the mass flow Rate of the refrigerant we must

Use the formula for a heat exchanger between the chiller with the refrigerant and the natural gas.

$$\dot{m}_1 (H_4 - H_1) = \dot{m}_5 (H_6 - H_5)$$

The enthalpies H_1, H_5, H_6 can be calculated using the HIG() function and the H_4 can be calculated using the formula $H_{LSAT} = H_V - \Delta H_{vap}$, where H_V is found using the HIG() function and ΔH_{vap} can be found using the HVAP() function. Using the HIG() and HVAP() function we get. $H_1 = -108364, H_4 = -117027, H_5 = -74901, H_6 = -76998$ Where all enthalpies have units of (kJ/kmol)

$$\dot{m}_5 = 100 \text{ MMSCFD}$$

In order to convert 100 MMSCFD to a mass flow rate you must use the ideal gas equation.

$$1.) PV = NRT$$

$$2.) V = NRT/P$$

In order to use the ideal gas law for English units you must use the English constant values: ($R = \frac{10.7239(\text{psia} \cdot \text{ft}^3)}{\text{lbmol} \cdot \text{R}}$, $T = 519.67^\circ \text{R}$, $P = 14.696(\text{psia})$)

$$3.) (10^8)(\text{SCFD}) = \left(N * \frac{\left(\frac{10.7239(\text{psia} \cdot \text{ft}^3)}{\text{lbmol} \cdot \text{R}} \right) * 519.67^\circ \text{R}}{14.696(\text{psia})} \right)$$

In equation (3) volume is equal to one hundred million standard cubic feet per day

$$4.) N = 263727.3623 \left(\frac{\text{lbmol}}{\text{day}} \right) * (16.04) \left(\frac{\text{lb}}{\text{lbmol}} \right) = 4230186.891 \left(\frac{\text{lb}}{\text{day}} \right)$$

$$5.) 4230186.891 \left(\frac{\text{lb}}{\text{day}} \right) * \left(\frac{2.2\text{kg}}{\text{lb}} \right) = 1918780.4974316 \left(\frac{\text{Kg}}{\text{day}} \right)$$

$$6.) 1918780.4974316 \left(\frac{\text{Kg}}{\text{day}} \right) * \left(\frac{1\text{day}}{24\text{hr}} \right) * \left(\frac{1\text{hr}}{3600\text{s}} \right) = 1.38 \left(\frac{\text{Kmol}}{\text{s}} \right)$$

$$7.) \dot{m}_5 = 1.38 \left(\frac{\text{Kmol}}{\text{s}} \right)$$

with all the components of the heat exchanger equation solved for except for $\dot{m}_1$ we can now solve for $\dot{m}_1$

$$1.) \dot{m}_1 = - \left(\dot{m}_5 \left(\frac{\text{Kmol}}{\text{s}} \right) * (H_6 - H_5) \right) \left(\frac{\text{kJ}}{\text{kmol}} \right) / (H_4 - H_1) (\text{kJ/kmol})$$

$$2.) \dot{m}_1 = 1.38 \left(\frac{\text{Kmol}}{\text{s}} \right) * \frac{(-76998 - -74901) \left(\frac{\text{kJ}}{\text{kmol}} \right)}{(-117027 - -108364) \left(\frac{\text{kJ}}{\text{kmol}} \right)}$$

$$3.) \dot{m}_1 = .335 \left(\frac{\text{Kmol}}{\text{s}} \right)$$

Now that we have $\dot{m}_1$ we can calculate the amount of refrigerant required in KMTA (Assuming a 90% operating factor)

$$1.) \dot{m}_1 = .335 \left(\frac{\text{Kmol}}{\text{s}} \right) * 44.1 \left(\frac{\text{kg propane}}{\text{kmol}} \right) = 14.7735 \left(\frac{\text{kg}}{\text{s}} \right)$$

$$2.) 14.7735 \left(\frac{\text{kg}}{\text{s}} \right) * 3600 \left(\frac{\text{s}}{\text{hr}} \right) * 24 \left(\frac{\text{hr}}{\text{day}} \right) * 365 \left(\frac{\text{days}}{\text{year}} \right) = 465897096 \left(\frac{\text{kg}}{\text{year}} \right)$$

$$3.) 465897096 \left(\frac{\text{kg}}{\text{year}} \right) * \left(\frac{1\text{tonne}}{1000\text{kg}} \right) = 465897.096 \left(\frac{\text{tonnes}}{\text{year}} \right)$$

$$4.) 465897.096 \left(\frac{\text{tonnes}}{\text{year}} \right) * \left(\frac{1\text{kilo metric tonne}}{1000\text{ tonne}} \right) = 465.897\text{KMTA}$$

5) $465.897 \text{ KMTA} * (.9) = 419.3 \text{ KMTA}$ assuming a 90% operating factor.

3 Part(2)

The duties of the chiller and condenser:

To calculate the duty of the chiller we must use the equation

$$W_{chiller} = \dot{m}_{ref} * (H_4 - H_1)$$

H_1 can be obtained using the HIG() because it is an ideal gas.

H_4 is a saturated liquid and therefore can be obtained when applying the equation

$$H_{LSAT} = H_V - \Delta H_{vap}$$

The mass flow rate was found earlier in part one and was found to be $.335 \left(\frac{\text{Kmol}}{\text{s}} \right)$

The duty of the Chiller can now be calculated

$$1) W_{chiller} = .335 \left(\frac{\text{Kmol}}{\text{s}} \right) * (-117027 - -108364) \left(\frac{\text{kJ}}{\text{kmol}} \right)$$

$$2) W_{chiller} = 2901.6905 (\text{kJ/s})$$

$$3) W_{chiller} = 2.9016905 \text{ MW}$$

To find the duty of the condenser we must use the equation

$$W_{condenser} = \dot{m}_{ref} * (H_3 - H_2)$$

The state of H_3 is given to us and is a saturated liquid therefore we must use the equation below to solve for H_3 :

$$H_{LSAT} = H_V - \Delta H_{vap}$$

H_2 can be calculated using the HIG() function because it is an ideal gas.

The mass flow rate was found earlier in part one and was found to be

$$.335 \left(\frac{\text{Kmol}}{\text{s}} \right)$$

The duty of the condenser is can now be calculated:

$$1)W_{condenser\ ideal} = .335 \left(\frac{kmol}{s} \right) (-117027 - -102867) \left(\frac{kJ}{kmol} \right)$$

$$2)W_{condenser\ ideal} = -4743.2327 \left(\frac{kJ}{s} \right)$$

$$3)W_{condenser\ ideal} = -4.743.2327\ MW$$

$$1)W_{condenser\ polytropic} = .3015 \left(\frac{kmol}{s} \right) (-117027 - -100784) \left(\frac{kJ}{kmol} \right)$$

$$2)W_{condenser\ polytropic} = -4896.7120 \left(\frac{kJ}{s} \right)$$

$$3)W_{condenser\ polytropic} = -4.8967120\ MW$$

$$1)W_{condenser\ isentropic} = .3015 \left(\frac{kmol}{s} \right) (-117027 - -102867) \left(\frac{kJ}{kmol} \right)$$

$$2)W_{condenser\ isentropic} = -4821.3721 \left(\frac{kJ}{s} \right)$$

$$3)W_{condenser\ isentropic} = -4.8213721\ MW$$

3 Part(3)

The Power requirement of the compressor is calculated by

$$W_{compressor\ isentropic} = \dot{m}_{ref} * (H_{2\ isentropic} - H_{1\ isentropic})$$

The $H_{2\ isentropic}$ take into consideration the .75 efficiency and there is no need to divide the work by .75 again

$$W_{compressor\ isentropic} = .3015 \left(\frac{Kmol}{s} \right) * (-101034 - -108364)$$

$$W_{compressor\ isentropic} = 2209.8506 \left(\frac{kJ}{s} \right)$$

$$W_{compressor\ isentropic} = 2.209\ MW$$

$$W_{Polytropic} = \frac{\gamma}{\gamma-1} RT \left[\left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} - 1 \right]$$

$$\gamma = \frac{C_P}{C_V}$$

The values I acquired from online(engineering toolbox) was $C_P = .0634 \left(\frac{kJ}{kmol} \right)$ and $C_V = .0536 \left(\frac{kJ}{kmol} \right)$ at -42°C

$$\gamma = \frac{.0634 \left(\frac{kJ}{kmol} \right)}{.0536 \left(\frac{kJ}{kmol} \right)}$$

$$\gamma = 1.1822$$

$$P_2 = 11.4361 \text{ bar}$$

$$P_1 = 1 \text{ bar}$$

$$R = .1886 \left(\frac{kJ}{kgK} \right)$$

$$W_{Polytropic} = 128.1954 \left(\frac{kJ}{kg} \right) * 44.1 \left(\frac{kJ}{kmol} \right) * .3015 \left(\frac{Kmol}{s} \right) / .75$$

$$W_{Polytropic} = 2.285 \text{ MW}$$

3 Part(4)

The COP for a refrigeration cycle can be calculated using the following formula

$$\text{COP} = \frac{Q_C}{Q_H - Q_C}$$

The COP can be calculated with the isentropic efficiency and the polytropic efficiency. The same formula is applied to each of them. The

Q_H value will not change from the ideal because the only thing happening at the chiller is the removal of latent heat and the chiller is adiabatic. however the Q_C value will change depending on the different types of efficiency because it is going through a condenser which yields different real enthalpies. Each Q_H :

$$1) Q_H = \dot{m} * (H_1 - H_4)$$

$$Q_H = .3015 \left(\frac{\text{Kmol}}{\text{s}} \right) * (-117027 - -108364) \left(\frac{\text{kJ}}{\text{kmol}} \right)$$

$$Q_H = 2611.15$$

The $Q_{C \text{ isentropic}}$ value is equivalent to the isentropic duty of the condenser and the $Q_{H \text{ isentropic}}$ value is equivalent to the isentropic duty of the condenser.

$$Q_{C \text{ isentropic}} = -4743.2327$$

$$Q_{C \text{ polytropic}} = -4896.7120$$

the two different Q_C values Result in two real COP values where the Q_H and Q_C inputted in equation 1 are the absolute values.

$$COP_{\text{isentropic}} \text{ and } COP_{\text{polytropic}}$$

$$COP_{\text{isentropic}} = 2.1818$$

$$COP_{\text{polytropic}} = 2.1428$$

3 Part(5)

The maximum COP for the refrigeration cycle can be calculated with the formula:

$$COP_{\text{max}} = \frac{T_C}{T_H - T_C}$$

$$T_C = 230 \text{ kelvin}$$

$$T_H = 312 \text{ kelvin}$$

$$COP_{max} = \frac{230K}{312K - 230K}$$

$$COP_{max} = 2.8492$$